

Gadd45a promotes DNA demethylation through TDG.

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Growth arrest and DNA-damage-inducible protein 45 (Gadd45) family members have been implicated in DNA demethylation in vertebrates. However, it remained unclear how they contribute to the demethylation process. Here, we demonstrate that Gadd45a promotes active DNA demethylation through thymine DNA glycosylase (TDG) which has recently been shown to excise 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC) generated in Ten-eleven-translocation (Tet)-initiated oxidative demethylation. The connection of Gadd45a with oxidative demethylation is evidenced by the enhanced activation of a methylated reporter gene in HEK293T cells expressing Gadd45a in combination with catalytically active TDG and Tet. Gadd45a interacts with TDG physically and increases the removal of 5fC and 5caC from genomic and transfected plasmid DNA by TDG. Knockout of both Gadd45a and Gadd45b from mouse ES cells leads to hypermethylation of specific genomic loci most of which are also targets of TDG and show 5fC enrichment in TDG-deficient cells. These observations indicate that the demethylation effect of Gadd45a is mediated by TDG activity. This finding thus unites Gadd45a with the recently defined Tet-initiated demethylation pathway.

TET Family Members Are Integral to Porcine Oocyte Maturation and Parthenogenetic Pre-Implantation Embryogenesis.

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The ten-eleven translocation (TET) enzyme family, which includes TET1/2/3, participates in active DNA demethylation in the eukaryotic genome; moreover, TET1/2/3 are functionally redundant in mice embryos. However, the combined effect of TET1/2/3 triple-gene knockdown or knockout on the porcine oocytes or embryos is still unclear. In this study, using Bobcat339, a specific small-molecule inhibitor of the TET family, we explored the effects of TET enzymes on oocyte maturation and early embryogenesis in pigs. Our results revealed that Bobcat339 treatment blocked porcine oocyte maturation and triggered early apoptosis. Furthermore, in the Bobcat339-treated oocytes, spindle architecture and chromosome alignment were disrupted, probably due to the huge loss of 5-hydroxymethylcytosine (5hmC) and concurrent increase in 5-methylcytosine (5mC). After Bobcat339 treatment, early parthenogenetic embryos exhibited abnormal 5mC and 5hmC levels, which resulted in compromised cleavage and blastocyst rate. The mRNA levels of EIF1A and DPPA2 (ZGA marker genes) were significantly decreased, which may explain why the embryos were arrested at the 4-cell stage after Bobcat339 treatment. In addition, the mRNA levels of pluripotency-related genes OCT4 and NANOG were declined after Bobcat339 treatment. RNA sequencing analysis revealed differentially expressed genes in Bobcat339-treated embryos at the 4-cell stage, which were significantly enriched in cell proliferation, cell component related to mitochondrion, and cell adhesion molecule binding. Our results indicated that TET proteins are essential for porcine oocyte maturation and early embryogenesis, and they act by mediating 5mC/5hmC levels and gene transcription.

Ten-Eleven Translocation-3 CXXC domain is critical for postfertilization demethylation and expression of pluripotency genes in pig embryo†.

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Enzymes of the ten-eleven translocation family are considered to play an important role in the regulation of DNA methylation patterns by converting 5-methylcytosine to 5-hydroxymethylcytosine. Known as a maternal transcript enriched in mature oocytes, ten-eleven translocation-3 (TET3) has been suggested to initiate DNA demethylation of the paternal genome in zygotes. Previous studies in mouse cells indicate that the N-terminal CXXC domain of TET3 is important in catalyzing the oxidation of 5-methylcytosine through its potential DNA binding ability; however, it is not clear whether the DNA binding capacity of CXXC domain is required for the 5-hydroxymethylcytosine conversion in mammalian embryos. Here, we identified TET3 isoforms in porcine oocytes and investigated the role of the oocyte specific TET3 isoform (pTET3L) in controlling postfertilization demethylation in porcine embryos. The pTET3L possessed sequences representing a known DNA binding domain, the CXXC, and injection of the TET3 CXXC fused with GFP into mature porcine oocytes resulted in exclusive localization of the GFP-CXXC in the pronuclei. The CXXC overexpression reduced the 5-methylcytosine level in zygotes and enhanced the DNA demethylation of the NANOG promoter in 2-cell stage embryos. Furthermore, there was an increase in the transcript abundance of NANOG and ESRRB in blastocysts developed from GFP-CXXC injected oocytes. Targeted knockdown of pTET3L resulted in the downregulation of pluripotency genes in subsequently developed blastocysts. The findings indicate that the CXXC domain of TET3 serves as a critical component for the postfertilization demethylation of porcine embryos and coordinates proper expression of pluripotency related genes in blastocysts.

Hydroxylation of 5-methylcytosine by TET1 promotes active DNA demethylation in the adult brain.

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Cytosine methylation is the major covalent modification of mammalian genomic DNA and plays important roles in transcriptional regulation. The molecular mechanism underlying the enzymatic removal of this epigenetic mark, however, remains elusive. Here, we show that 5-methylcytosine (5mC) hydroxylase TET1, by converting 5mCs to 5-hydroxymethylcytosines (5hmCs), promotes DNA demethylation in mammalian cells through a process that requires the base excision repair pathway. Though expression of the 12 known human DNA glycosylases individually did not enhance removal of 5hmCs in mammalian cells, demethylation of both exogenously introduced and endogenous 5hmCs is promoted by the AID (activation-induced deaminase)/APOBEC (apolipoprotein B mRNA-editing enzyme complex) family of cytidine deaminases. Furthermore, Tet1 and Apobec1 are involved in neuronal activity-induced, region-specific, active DNA demethylation and subsequent gene expression in the dentate gyrus of the adult mouse brain in vivo. Our study suggests a TET1-induced oxidation-deamination mechanism for active DNA demethylation in mammals.

Tet Enzymes, Variants, and Differential Effects on Function.

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Discovery of the ten-eleven translocation 1 (TET) methylcytosine dioxygenase family of enzymes, nearly 10 years ago, heralded a major breakthrough in understanding the epigenetic modifications of DNA. Initially described as catalyzing the oxidation of methyl cytosine (5mC) to hydroxymethyl cytosine

(5hmC), it is now clear that these enzymes can also catalyze additional reactions leading to active DNA demethylation. The association of TET enzymes, as well as the 5hmC, with active regulatory regions of the genome has been studied extensively in embryonic stem cells, although these enzymes are expressed widely also in differentiated tissues. However, TET1 and TET3 are found as various isoforms, as a result of utilizing alternative regulatory regions in distinct tissues. Some of these isoforms, like TET2, lack the CXXC domain which probably has major implications on their recruitment to specific loci in the genome, while in certain contexts TET1 is seen paradoxically to repress transcription. In this review we bring together these novel aspects of the differential regulation of these Tet isoforms and the likely consequences on their activity.

SAR insights into TET2 catalytic domain inhibition: Synthesis of 2-Hydroxy-4-Methylene-pentanedicarboxylates.

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The TET (Ten-Eleven Translocation) dioxygenase enzyme family comprising 3 members, TET1-3, play key roles in DNA demethylation. These processes regulate transcription programs that determine cell lineage, survival, proliferation, and differentiation. The impetus for our investigations described here is derived from the need to develop illuminating small molecule probes for TET enzymes with cellular activity and specificity. The studies were done so in the context of the importance of TET2 in the hematopoietic system and the preponderance of loss of function somatic TET2 mutations in myeloid diseases. We have identified that 2-hydroxy-4-methylene-pentanedicarboxylic acid 2a reversibly competes with the co-substrate α -KG in the TET2 catalytic domain and inhibits the dioxygenase activity with an $IC_{50} = 11.0 \pm 0.9 \mu M$ at $10 \mu M$ α -KG in a cell free system and binds in the TET2 catalytic domain with $K_d = 0.3 \pm 0.12 \mu M$.

Ten eleven translocation enzymes and 5-hydroxymethylation in mammalian development and cancer.

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5-Hydroxymethylcytosine (5hmC) is an oxidative product of 5-methylcytosine (5mC), catalyzed by the ten eleven translocation (TET) family of enzymes. Although 5hmC was discovered several decades ago, it was only after its recent identification in murine brain and stem cell DNA that it has become a major focus of epigenomic research. Part of the reason for this delay is due to the difficulty in detecting both global and locus-specific 5hmC levels. Several studies have addressed this issue with the development of novel techniques to locate and measure 5hmC, which led to multiple reports detailing 5hmC patterns in stem cells and global 5hmC levels during embryogenesis. Based on these studies of 5hmC levels and reports of tissue-specific TET expression, these enzymes are thought to play a role in mammalian development and differentiation. In addition, the TET enzymes are mutated in several types of cancer, affecting their activity and likely altering genomic 5hmC and 5mC patterns. Furthermore, oxidation of 5mC appears to be a step in several active DNA demethylation pathways, which may be important for normal processes, as well as global hypomethylation during cancer development and progression. Much has been revealed about this interesting DNA modification in recent years, but more research is needed for understanding the role of TET proteins and 5hmC in gene regulation and disease.

A highly sensitive signal-on biosensor based on restriction enzyme-mediated molecular switch for detection of TET1.

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Ten-eleven translocation 1 (TET1) is a member of the TET enzyme family of dioxygenases, which plays an important role in active DNA demethylation. Therefore, the sensitive TET1 detection could help us better understand DNA methylation-demethylation in epigenetics. Here we report a detection method that consists of electrode fabrication, TET1 modification, DNA digestion, signal-on oxidoreduction, and current peak monitoring. An exquisitely designed 5'-end-G-rich oligodeoxynucleotide was synthesized bearing a methylated cytosine (5-mC), which formed into hairpin dsDNA with the MspI recognition sequence (CmCGG/GGCC). Then hairpin dsDNA was fabricated onto gold nanoparticles modified glassy carbon electrode (DNA/AuNPs/GCE) via Au-S bond. The combination uses of restriction enzyme MspI and hemin converted fabricated-dsDNA into peroxidase-mimicking DNAzyme, thereby promoting the reduction of H₂O₂ with a current peak. However, the current peak was extremely decreased once TET1 and T4 β-GT were used in advance. We confirmed a delicately linear relationship matching between the current difference and TET1 activity from 0.7 to 10.5 ng μL⁻¹ with a detection limit of 0.027 ng μL⁻¹, which outcompeted the former methods at least one order of magnitudes. The TET1 activity evaluation in the existence of Bobcat339 was also tested as the proof of concept of inhibitors screening. Our strategy provides a novel, label-free, and sensitive electrochemical approach that enables us to complete both TET1 activity evaluation and potential TET1 inhibitors screening.

Excision of 5-Carboxylcytosine by Thymine DNA Glycosylase.

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5-Methylcytosine (mC) is an epigenetic mark that is written by methyltransferases, erased through passive and active mechanisms, and impacts transcription, development, diseases including cancer, and aging. Active DNA demethylation involves TET-mediated stepwise oxidation of mC to 5-hydroxymethylcytosine, 5-formylcytosine (fC), or 5-carboxylcytosine (caC), excision of fC or caC by thymine DNA glycosylase (TDG), and subsequent base excision repair. Many elements of this essential process are poorly defined, including TDG excision of caC. To address this problem, we solved high-resolution structures of human TDG bound to DNA with cadC (5-carboxyl-2'-deoxycytidine) flipped into its active site. The structures unveil detailed enzyme-substrate interactions that mediate recognition and removal of caC, many involving water molecules. Importantly, two water molecules contact a carboxylate oxygen of caC and are poised to facilitate acid-catalyzed caC excision. Moreover, a substrate-dependent conformational change in TDG modulates the hydrogen bond interactions for one of these waters, enabling productive interaction with caC. An Asn residue (N191) that is critical for caC excision is found to contact N3 and N4 of caC, suggesting a mechanism for acid-catalyzed base excision that features an N3-protonated form of caC but would be ineffective for C, mC, or hmC. We also investigated another Asn residue (N140) that is catalytically essential and strictly conserved in the TDG-MUG enzyme family. A structure of N140A-TDG bound to cadC DNA provides the first high-resolution insight into how enzyme-substrate interactions, including water molecules, are impacted by depleting the conserved Asn, informing its role in binding and addition of the nucleophilic water molecule.

The role of active DNA demethylation and Tet enzyme function in memory formation and cocaine action.

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Active DNA modification is a major epigenetic mechanism that regulates gene expression in an experience-dependent manner, which is thought to establish stable changes in neuronal function and behavior. Recent discoveries regarding the Ten eleven translocation (Tet1-3) family of DNA hydroxylases have provided a new avenue for the study of active DNA demethylation, and may thus help to advance our understanding of how dynamic DNA modifications lead to long-lasting changes in brain regions underlying learning and memory, as well as drug-seeking and propensity for relapse following abstinence. Drug addiction is a complex, relapsing disorder in which compulsive drug-seeking behavior can persist despite aversive consequences. Therefore, understanding the molecular mechanisms that underlie the onset and persistence of drug addiction, as well as the pronounced propensity for relapse observed in addicts, is necessary for the development of selective treatments and therapies. In this mini-review, we provide an overview of the involvement of active DNA demethylation with an emphasis on the Tet family of enzymes and 5-hydroxymethylcytosine (5-hmC) in learning and memory, as well as in drug-seeking behavior. Memory and addiction share overlapping molecular, cellular, and circuit functions allowing research in one area to inform the other. Current discrepancies and directions for future studies focusing on the dynamic interplay between DNA methylation and demethylation, and how they orchestrate gene expression required for neuronal plasticity underlying memory formation, are discussed.
